

Punto 1

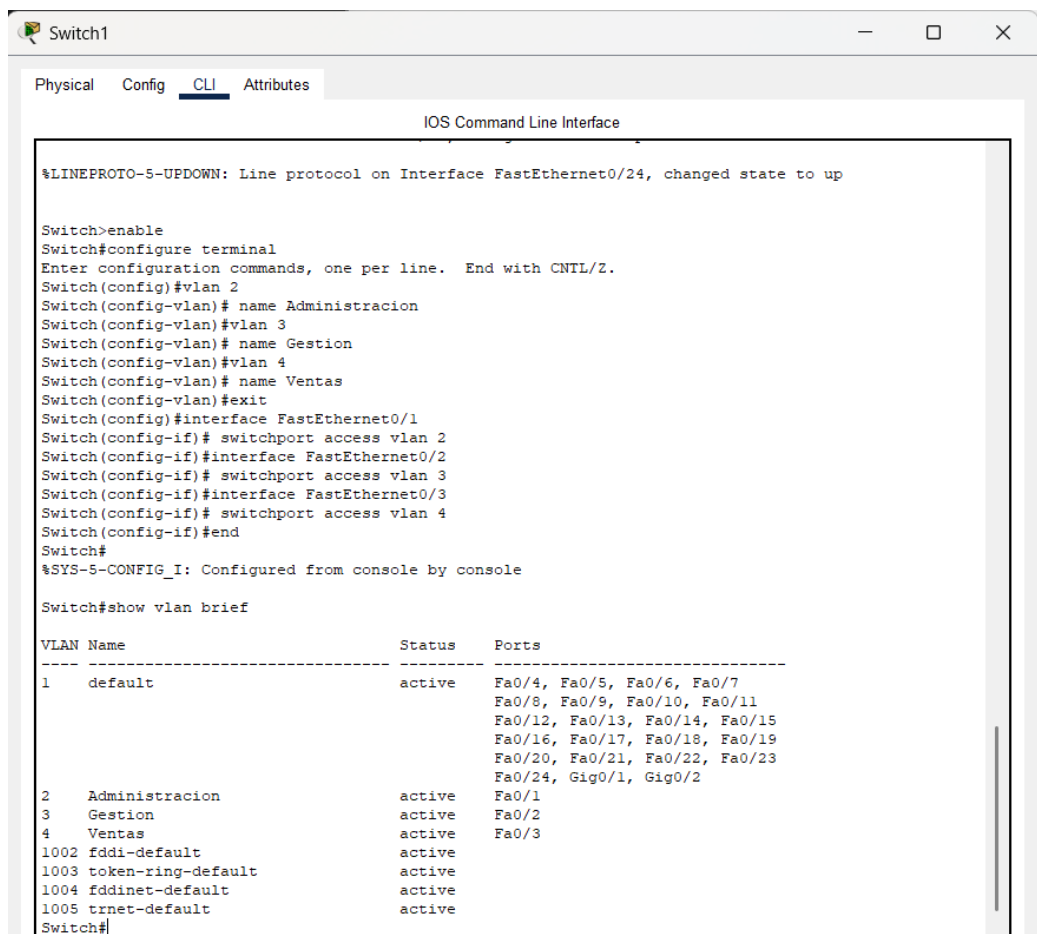
Ping desde una PC hacia el resto en la red plana. ¿Qué resultado obtienen?

Responden todas (0% de pérdida). Están en el mismo segmento 10.0.0.0/8 y en un único dominio de broadcast, por lo que hay conectividad total.

Punto 2

Con show vlan, ¿qué observamos en los puertos? ¿Están o no en el mismo segmento?

Cada puerto quedó asignado a su VLAN. A nivel IP siguen en el mismo segmento (todas 10.x/8), pero en capa 2 quedaron separadas en dominios de broadcast distintos: comparten direccionamiento pero las VLANs las aíslan.



```
Switch1
Physical Config CLI Attributes
IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)# name Administracion
Switch(config-vlan)#vlan 3
Switch(config-vlan)# name Gestion
Switch(config-vlan)#vlan 4
Switch(config-vlan)# name Ventas
Switch(config-vlan)#exit
Switch(config)#interface FastEthernet0/1
Switch(config-if)# switchport access vlan 2
Switch(config-if)#interface FastEthernet0/2
Switch(config-if)# switchport access vlan 3
Switch(config-if)#interface FastEthernet0/3
Switch(config-if)# switchport access vlan 4
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                           Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                           Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                           Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                           Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                           Fa0/24, Gig0/1, Gig0/2
2    Administracion         active    Fa0/1
3    Gestion                active    Fa0/2
4    Ventas                 active    Fa0/3
1002 fddi-default         active
1003 token-ring-default   active
1004 fddinet-default      active
1005 trnet-default        active
Switch#
```

show vlan brief: Fa0/1→VLAN2, Fa0/2→VLAN3, Fa0/3→VLAN4.

Punto 3

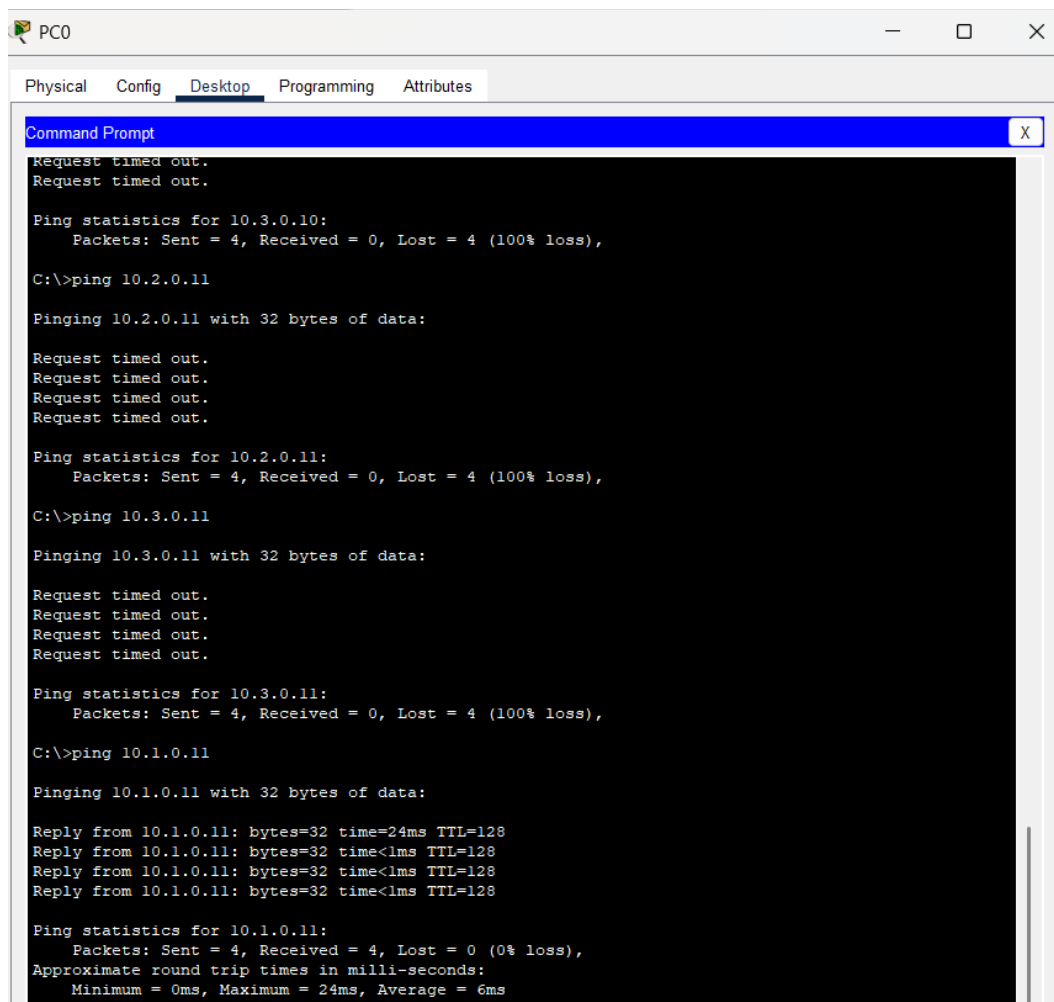
Tras asignar las VLANs, ¿qué pasa con el ping?

Falla (100%). Las PCs de distinta VLAN no se ven, y las de la misma VLAN en distinto switch tampoco, porque el enlace entre switches sigue en VLAN 1.

Punto 4

Ping desde 10.1.0.10 hacia el resto. ¿Qué pasa y por qué?

Tras pasar el enlace entre switches a VLAN 2, desde 10.1.0.10 solo se alcanza a 10.1.0.11 (su misma VLAN). No llega a VLAN 3 ni 4: cada VLAN es un dominio de broadcast independiente y el tráfico no cruza entre VLANs en capa 2.



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Request timed out.
Request timed out.

Ping statistics for 10.3.0.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.2.0.11

Pinging 10.2.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.2.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.1.0.11

Pinging 10.1.0.11 with 32 bytes of data:

Reply from 10.1.0.11: bytes=32 time=24ms TTL=128
Reply from 10.1.0.11: bytes=32 time<1ms TTL=128
Reply from 10.1.0.11: bytes=32 time<1ms TTL=128
Reply from 10.1.0.11: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 6ms
```

10.1.0.11 responde (misma VLAN); 10.2.0.11 y 10.3.0.11 fallan.

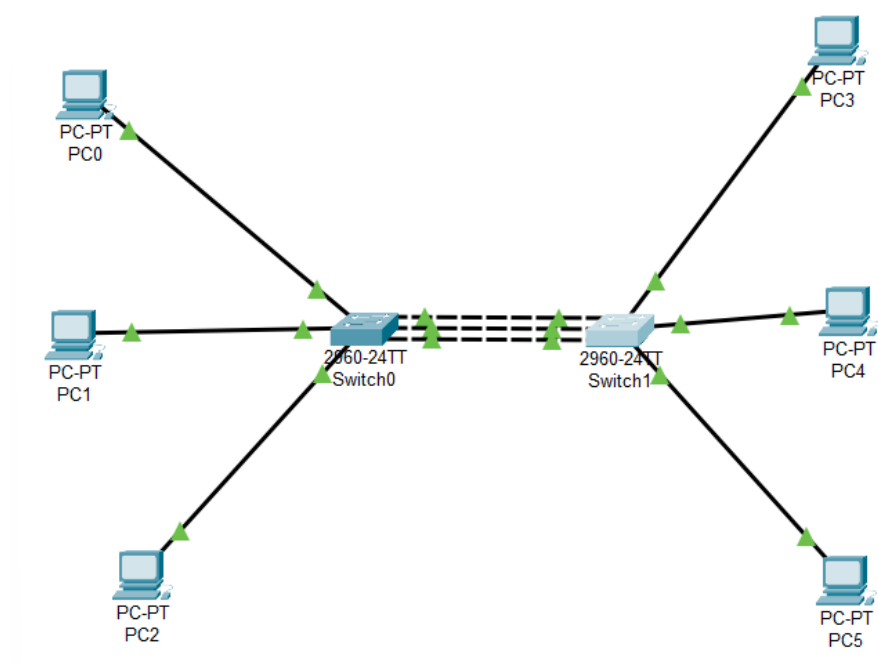
¿Formas de interconectar las VLANs entre switches?

Física: un cable por VLAN (cada puerto access en su VLAN). Lógica: un único trunk 802.1Q que transporta todas las VLANs etiquetadas.

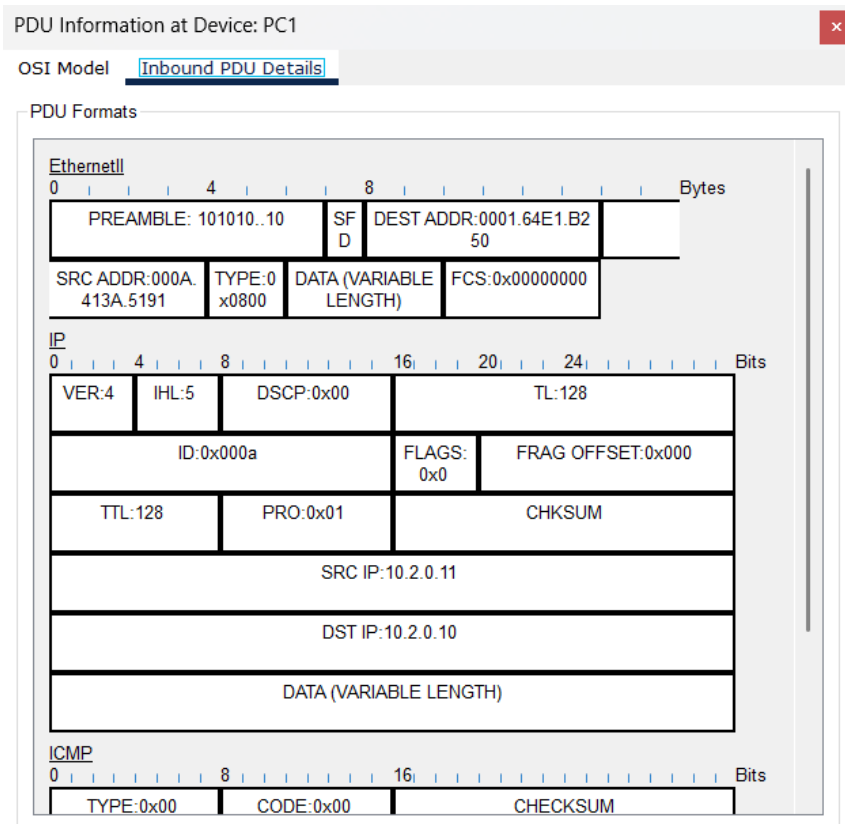
Punto 5 — Interconexión física

En simulación, ICMP entre 2 PCs de la misma VLAN: ¿se modifican las cabeceras?

No. Los enlaces son access y transportan las tramas sin etiquetar: la PDU muestra un EthernetII normal, sin campo 802.1Q.



Un cable por VLAN entre los switches.



PDU en la PC: EthernetII sin etiqueta 802.1Q (enlace access).

Punto 6 — Interconexión lógica (trunk 802.1Q)

Con el trunk, en simulación: ¿se modifican las cabeceras?

Sí, en el tramo del trunk: la trama recibe una etiqueta 802.1Q (VLAN ID) al entrar al trunk y se le quita al salir hacia el puerto access. La encapsulación 802.1q se ve en el show interfaces trunk.

¿Se puede acceder con ICMP entre VLANs?

No. Siguen siendo dominios de capa 2 separados; hace falta ruteo de capa 3.

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-3-UPDOWN: Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/24, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface FastEthernet0/24
Switch(config-if)# switchport mode trunk

Switch(config-if)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

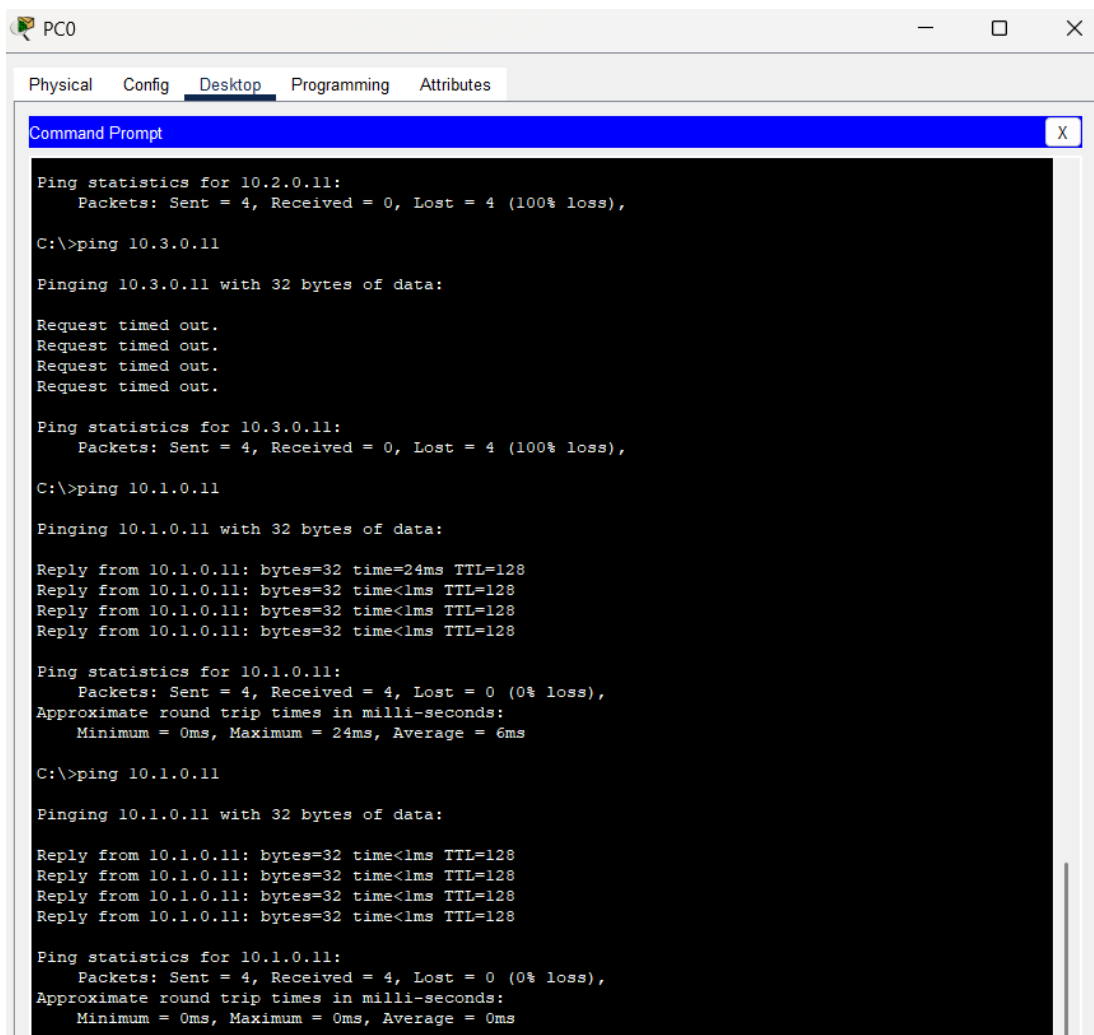
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Fa0/24    on        802.1q         trunking    1

Port      Vlans allowed on trunk
Fa0/24    1-1005

Port      Vlans allowed and active in management domain
Fa0/24    1,2,3,4

Port      Vlans in spanning tree forwarding state and not pruned
Fa0/24    1,2,3,4
```

show interfaces trunk: Fa0/24 trunking, encapsulación 802.1q, VLANs 1,2,3,4.



The screenshot shows a virtual PC named PC0 with a desktop environment. A Command Prompt window is open, displaying the results of several ping commands. The first ping is to 10.2.0.11, which fails with 100% loss. The second ping is to 10.3.0.11, which also fails with 100% loss. The third ping is to 10.1.0.11, which succeeds with 0% loss and a 6ms average round trip time. The fourth ping is also to 10.1.0.11 and succeeds with 0% loss and 0ms average round trip time.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Ping statistics for 10.2.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 10.3.0.11:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.1.0.11

Pinging 10.1.0.11 with 32 bytes of data:

Reply from 10.1.0.11: bytes=32 time=24ms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128

Ping statistics for 10.1.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 24ms, Average = 6ms

C:\>ping 10.1.0.11

Pinging 10.1.0.11 with 32 bytes of data:

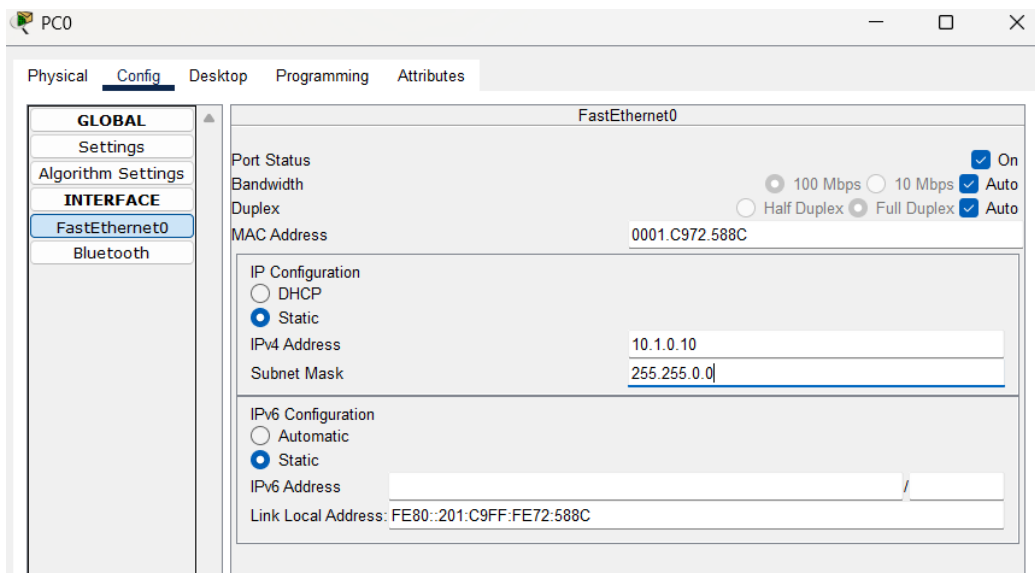
Reply from 10.1.0.11: bytes=32 time<lms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128
Reply from 10.1.0.11: bytes=32 time<lms TTL=128

Ping statistics for 10.1.0.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

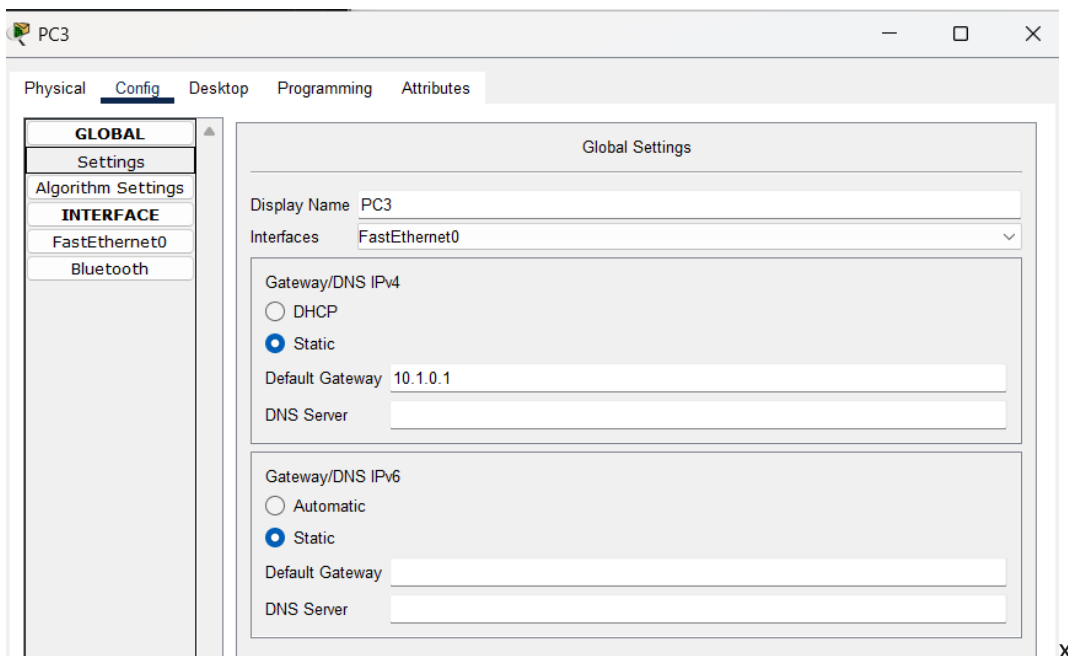
Por el trunk, PC0 (VLAN 2) llega a 10.1.0.11; el tráfico a otras VLANs sigue fallando.

Ruteo entre VLANs (/8 → /16)

Para comunicar las VLANs se cambia la máscara a /16 (255.255.0.0) y se asigna a cada PC el gateway de su VLAN: VLAN 2 → 10.1.0.1, VLAN 3 → 10.2.0.1, VLAN 4 → 10.3.0.1.



Ejemplo: PC0 con máscara 255.255.0.0.



Ejemplo: gateway de la VLAN (PC3 → 10.1.0.1).

Punto 7 — Ruteo con una interfaz por VLAN

Configure y repita la prueba de ICMP entre las diferentes PCs.

Con el router (una interfaz en cada VLAN) el ICMP entre distintas VLANs ya funciona. En las respuestas se ve TTL=127: el paquete pasó por el router.

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]:

Press RETURN to get started!

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)# ip address 10.1.0.1 255.255.0.0
Router(config-if)# no shutdown

Router(config-if)#interface FastEthernet1/0
Router(config-if)# ip address 10.2.0.1 255.255.0.0
Router(config-if)# no shutdown

Router(config-if)#interface FastEthernet6/0
Router(config-if)# ip address 10.3.0.1 255.255.0.0
Router(config-if)# no shutdown

Router(config-if)#end
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet6/0, changed state to up

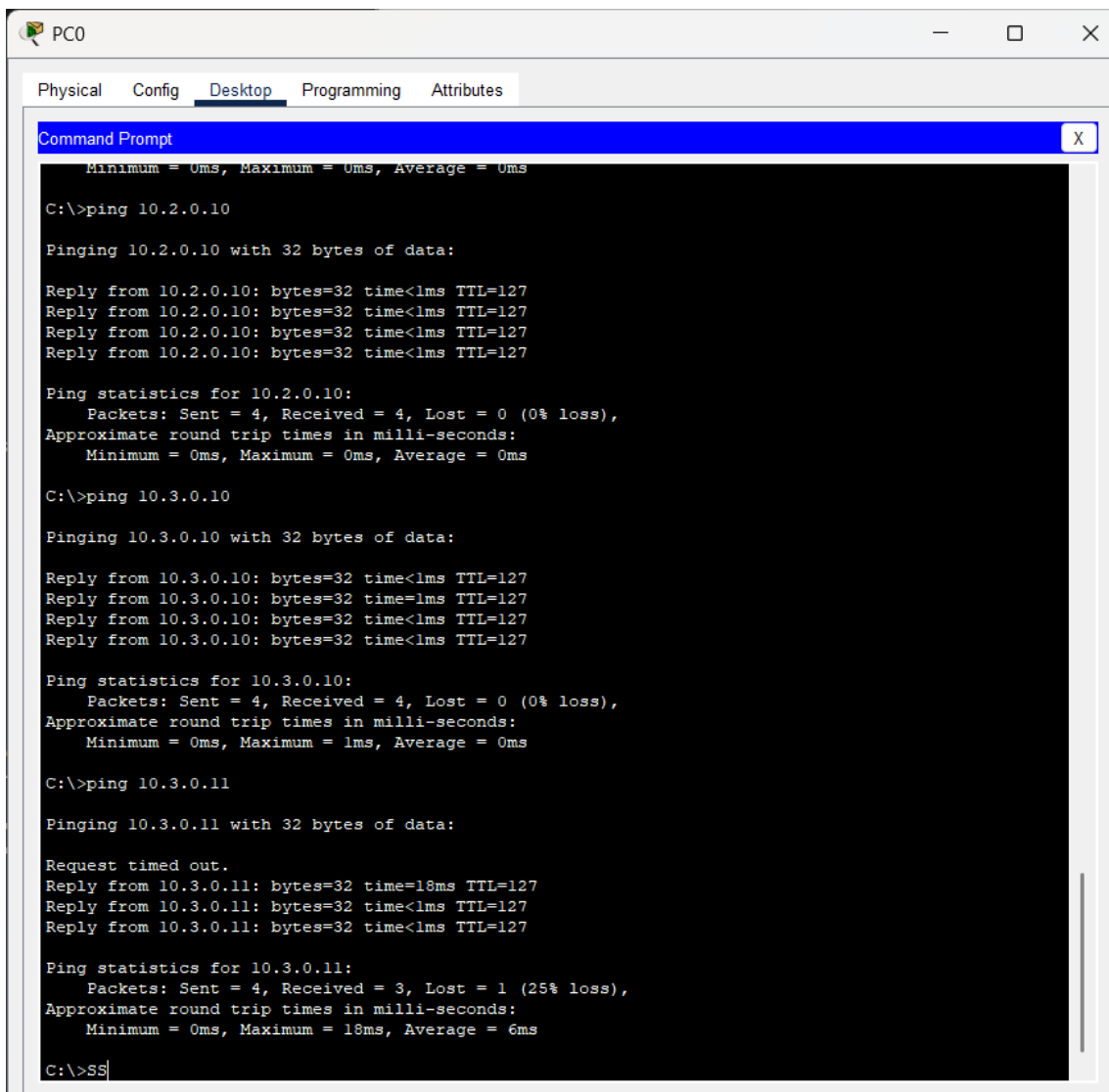
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet6/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

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Router: Fa0/0=10.1.0.1, Fa1/0=10.2.0.1, Fa6/0=10.3.0.1 (una interfaz por VLAN).



The screenshot shows a window titled 'PC0' with tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes'. The 'Desktop' tab is active, displaying a 'Command Prompt' window. The Command Prompt shows the following output:

```
Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.2.0.10

Pinging 10.2.0.10 with 32 bytes of data:

Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.2.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.3.0.10

Pinging 10.3.0.10 with 32 bytes of data:

Reply from 10.3.0.10: bytes=32 time<1ms TTL=127
Reply from 10.3.0.10: bytes=32 time<1ms TTL=127
Reply from 10.3.0.10: bytes=32 time<1ms TTL=127
Reply from 10.3.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.3.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.3.0.11

Pinging 10.3.0.11 with 32 bytes of data:

Request timed out.
Reply from 10.3.0.11: bytes=32 time=18ms TTL=127
Reply from 10.3.0.11: bytes=32 time<1ms TTL=127
Reply from 10.3.0.11: bytes=32 time<1ms TTL=127

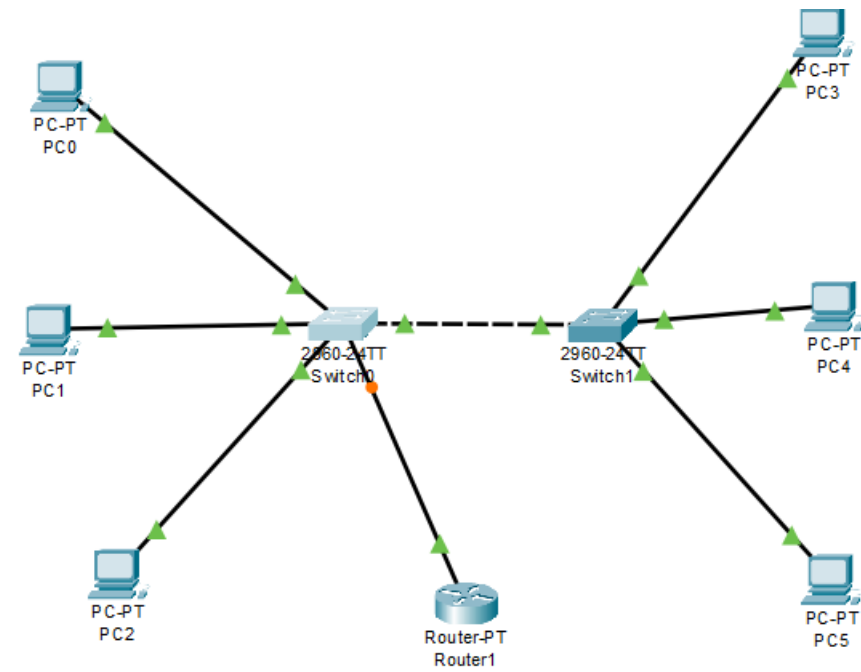
Ping statistics for 10.3.0.11:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 18ms, Average = 6ms

C:\>SS
```

PC0 (VLAN 2) → 10.2.0.10 (VLAN 3) y 10.3.0.10 (VLAN 4) responden con TTL=127.

Punto 8 — ¿Es escalable a 20 VLANs?

No. Una interfaz física por VLAN no escala (20 VLANs = 20 interfaces). La solución escalable es router-on-a-stick: un solo enlace trunk entre switch y router, con subinterfaces etiquetadas 802.1Q (una por VLAN). Con esto el ruteo entre las 3 VLANs funciona usando un único cable.



Router conectado al switch con un solo cable (trunk).

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.4, changed state to up
%SYS-5-CONFIG_I: Configured from console by console

Router#
Router#enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet1/0
Router(config-if)# no ip address
Router(config-if)# shutdown

Router(config-if)#interface FastEthernet6/0
Router(config-if)# no ip address
Router(config-if)# shutdown

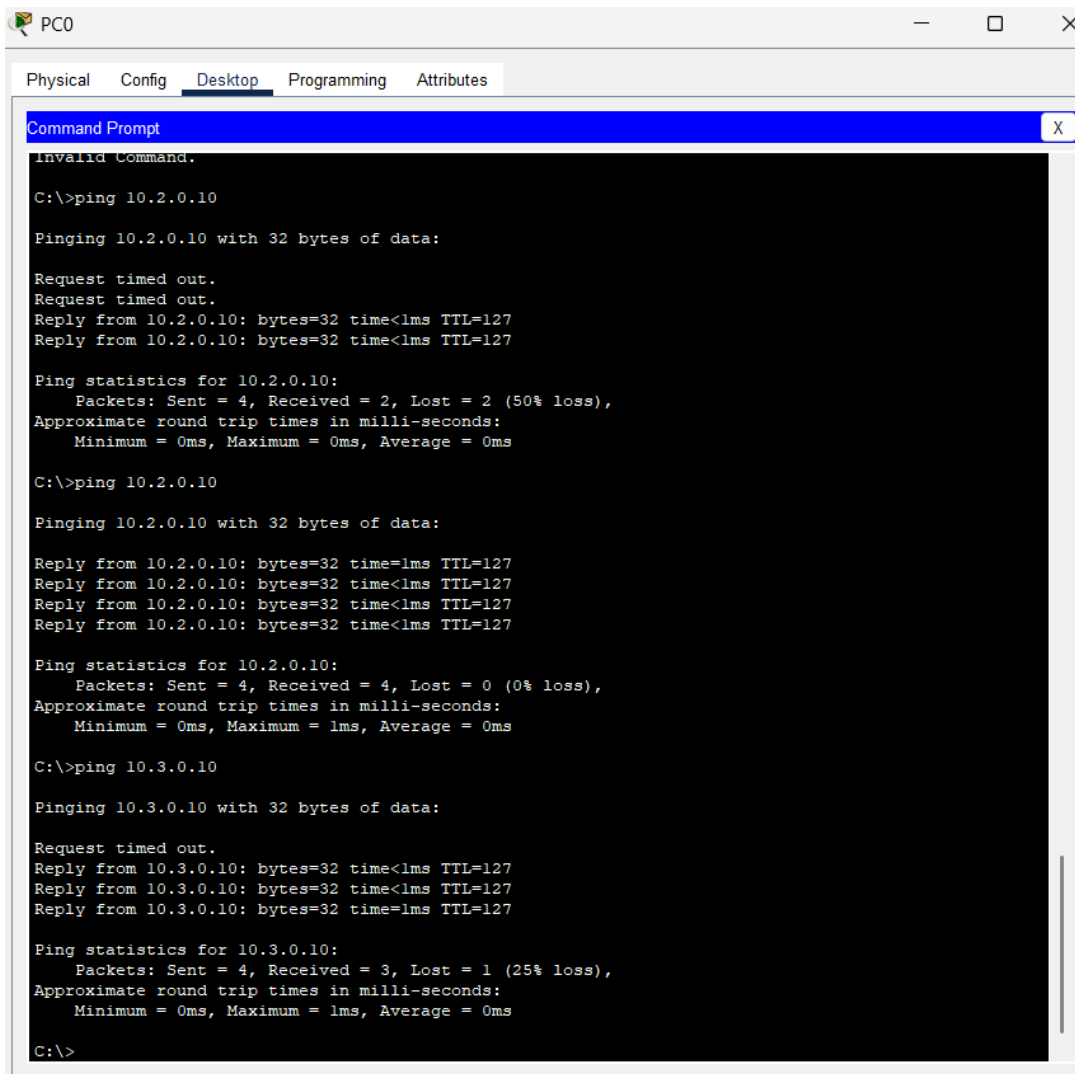
Router(config-if)#interface FastEthernet0/0.3
Router(config-subif)# ip address 10.2.0.1 255.255.0.0
Router(config-subif)#interface FastEthernet0/0.4
Router(config-subif)# ip address 10.3.0.1 255.255.0.0
Router(config-subif)#end
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet6/0, changed state to administratively down

Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface          IP-Address      OK? Method Status          Protocol
FastEthernet0/0    unassigned      YES manual up              up
FastEthernet0/0.2  10.1.0.1        YES manual up              up
FastEthernet0/0.3  10.2.0.1        YES manual up              up
FastEthernet0/0.4  10.3.0.1        YES manual up              up
FastEthernet1/0    unassigned      YES manual administratively down down
Serial2/0          unassigned      YES unset  administratively down down
Serial3/0          unassigned      YES unset  administratively down down
FastEthernet4/0    unassigned      YES unset  administratively down down
FastEthernet5/0    unassigned      YES unset  administratively down down
FastEthernet6/0    unassigned      YES manual administratively down down
Router#
```

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Subinterfaces Fa0/0.2/.3/.4 (encapsulation dot1Q) en up/up.



The screenshot shows a PC0 window with a Command Prompt open. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with Desktop selected. The Command Prompt shows the following output:

```
Invalid Command.

C:\>ping 10.2.0.10

Pinging 10.2.0.10 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.2.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.2.0.10

Pinging 10.2.0.10 with 32 bytes of data:

Reply from 10.2.0.10: bytes=32 time=1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127
Reply from 10.2.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.2.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.3.0.10

Pinging 10.3.0.10 with 32 bytes of data:

Request timed out.
Reply from 10.3.0.10: bytes=32 time<1ms TTL=127
Reply from 10.3.0.10: bytes=32 time<1ms TTL=127
Reply from 10.3.0.10: bytes=32 time=1ms TTL=127

Ping statistics for 10.3.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Ruteo entre VLANs con un solo cable (TTL=127): router-on-a-stick, escalable.